Quantum Fault Tolerance Meets Toom's Contours

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1 Hadamard and S Gates

Claim 1.1. For any generator subset S' of size d', denote by $x_{S'}$ and $z_{S'}$ the following assignments:

$$x_{S'} := \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S'}} \quad z_{S'} := \sum_{u \in \langle S/S' \rangle} \theta_{\mathbf{u},\mathbf{S'}}$$

Then $x_{S'}$ and $z_{S'}$ are binary representations of anticommuting logical codewords of the (d, d')-Toric code. In particular, $x_{S'} \in \mathcal{C}_X/\mathcal{C}_Z^{\perp}$, $z_{S'} \in \mathcal{C}_Z/\mathcal{C}_X^{\perp}$, and $x_{S'}z_{S'} = 1$. In addition, let $X_{S'}$ and $Z_{S'}$ be the Pauli matrices obtained by multiplying each non-trivial coordinate of them by X or Z, respectively.

Proof. By definition $x_{S'}$ and $z_{S'}$ are binary assignments on the d'-dimensional faces. We start by showing that $x_{S'} \in \mathcal{C}_X$ and $z_{S'} \in \mathcal{C}_Z$:

$$\begin{aligned}
\partial^- \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S'}} &= \sum_{u \in \langle S' \rangle} \sum_{g \in S'} \theta_{\mathbf{u},\mathbf{S'/g}} + \theta_{\mathbf{gu},\mathbf{S'/g}} \\
&= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S'/g}} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{gu},\mathbf{S'/g}} \\
&= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S'/g}} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S'/g}} = 0
\end{aligned}$$

$$\begin{aligned}
\partial^+ \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S/S'}} &= \sum_{u \in \langle S' \rangle} \sum_{g \in S'} \theta_{\mathbf{u},\mathbf{S/S' \cup g}} + \theta_{\mathbf{g^{-1}u},\mathbf{S/S' \cup g}} \\
&= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S/S' \cup g}} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{g^{-1}u},\mathbf{S/S' \cup g}} \\
&= \sum_{g \in S'} \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S/S' \cup g}} + \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S/S' \cup g}} = 0
\end{aligned}$$

Thus $x_{S'} \in \mathcal{C}_X$ and $z_{S'} \in \mathcal{C}_Z$, now for showing that they are not binary representations of stabiliers, it's enough to show they admit no vanishing product.

$$x_{S'}z_{S'} = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S'}} \sum_{u \in \langle S/S' \rangle} \theta_{\mathbf{u},\mathbf{S'}} = \theta_{\mathbf{e},\mathbf{S'}} \theta_{\mathbf{e},\mathbf{S'}} = 1$$

□

Definition 1.2 (The Transpose Map)**.** Let d, d' be integers, and let $\mathcal{G}$ be a (d, d')-Toric complex generated by the generator set S. We define the *transpose map* $\phi : \mathcal{F}^{d'} \to \mathcal{F}^{d-d'}$ by acting on the basis elements as follows:

$$\phi(\theta_{\mathbf{v},\mathbf{S'}}) \to \theta_{\mathbf{v^{-1}},\mathbf{S/S'}}$$

Notice that for $d' = d - d'$, the transpose map is a permutation over the bits.

Definition 1.3 (Hadamard-Type Gate H_ϕ)**.** For $d' = d - d'$, we define the logical Hadamard-type gate H_ϕ: $L \to L$ as follows, for any $S' \subset S$, of size d':

$$\begin{aligned} H_\phi(X_{S'}) &\to Z_{S/S'} \\ H_\phi(Z_{S'}) &\to X_{S/S'} \end{aligned}$$

Claim 1.4. The Hadamard-type gate H_ϕ has a fold-transversal implementation by, first applying H^n over all the physical qubits, and then permuting them by the transpose map ϕ, namely $\hat{H}_\phi = \phi \circ H^n$.

Proof. Clearly, for any subset of generators $S' \subset S$, the map ϕ sends $x_{S'}$ to $z_{S/S'}$:

$$\phi(x_{S'}) = \phi(\sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S'}}) = \sum_{u \in \langle S' \rangle} \theta_{\mathbf{u},\mathbf{S/S'}} = z_{S/S'}$$

Moreover, ϕ sends binary representations of Z-stabilizers to binary representations of X-stabilizers, and vice versa. To see this, consider a $d' + 1$ dimensional face, and consider the image of the stabilizer it defines.

$$\begin{aligned} \phi\left(\partial^- \theta_{\mathbf{v},\mathbf{S'}}\right) &= \phi\left(\sum_{g \in S'} \theta_{\mathbf{v},\mathbf{S'/g}} + \theta_{\mathbf{gv},\mathbf{S'/g}}\right) \\ &= \sum_{g \in S'} \theta_{\mathbf{v^{-1}},(\mathbf{S/S'})\cup\mathbf{g}} + \theta_{\mathbf{g^{-1}v^{-1}},(\mathbf{S/S'})\cup\mathbf{g}} \\ &= \partial^+ \theta_{\mathbf{v^{-1}},\mathbf{S/S'}} \end{aligned}$$

And since $\phi \circ \phi = I$, we also have that $\phi\left(\partial^+ \theta_{\mathbf{v^{-1}},\mathbf{S/S'}}\right)$. Thus, $H^{\otimes n} \circ \phi$ sends Z-stabilizers to X-stabilizers and vice versa, thereby preserving the code space. Combining this with the above, $H^{\otimes n} \circ \phi$ realizes the logical form H_ϕ in the $(d, d/2)$-Toric code. $\square$

Corollary 1.5. For the $(d, d/2)$-Toric code, consider the gate $CZ_{S' \to S/S'}$: $L^{\otimes 2} \to L^{\otimes 2}$, defined to act on two encoded registers by applying $Z_{S/S'}$ on the second register, controlled by whether the logical $X_{S'}$ is turned on in the first register. Then $CZ_{S' \to S/S'}$ has a fold-transversal realization.

Proof. Consider the following circuit,

1. Apply H^n on the second register.
2. For any bit $\theta_{\mathbf{v},\mathbf{S'}}$, apply CX from $\theta_{\mathbf{v},\mathbf{S'}}$ in the first register into $\phi(\theta_{\mathbf{v},\mathbf{S'}})$ in the second register.

3. Again apply H^n on the second register.

$\square$

Definition 1.6 (The Nice Encoding)**.** Consider $(d, d/2)$-Toric code and fix a subset $S' \subset S$, at size d'. We define the *nice encoding* to be the subsystem code, which encodes a single logical qubit as follow:

$$
\begin{aligned}
\left|\bar{0}\right\rangle &= \left|\mathcal{C}_Z^{\perp}\right\rangle \\
\left|\bar{1}\right\rangle &= \left|x_{S'} + x_{S/S'} + \mathcal{C}_Z^{\perp}\right\rangle
\end{aligned}
$$

Claim 1.7. Given a resource state, the nice encoding has a fold-transversal realization of the logical $\mathbf{S}$ gate.

Proof. Consider the resource state defined by:

$$
\left|\mathbf{S}\right\rangle = \frac{1}{\sqrt{2}}\left(\left|\bar{0}\right\rangle + i\left|\bar{0} + x_{S'}\right\rangle\right)
$$

Figure 1: Simulating the $\mathbf{S}$-gate, using the $\left|\mathbf{S}\right\rangle$ magic state.

$$
\begin{aligned}
\left|\psi\right\rangle\left|\mathbf{S}\right\rangle &= \left(\alpha\left|\bar{0}\right\rangle + \beta\left|\bar{0} + x_{S'} + x_{S/S'}\right\rangle\right)\frac{1}{\sqrt{2}}\left(\left|\bar{0}\right\rangle + i\left|\bar{0} + x_{S'}\right\rangle\right) \\
&\mapsto \frac{1}{\sqrt{2}}\big(\alpha\left|\bar{0}\right\rangle\left|\bar{0}\right\rangle + i\alpha\left|\bar{0}\right\rangle\left|\bar{0} + x_{S'}\right\rangle \\
&\quad + \beta\left|\bar{0} + x_{S'} + x_{S/S'}\right\rangle\left|\bar{0} + x_{S'} + x_{S/S'}\right\rangle + i\beta\left|\bar{0} + x_{S'} + x_{S/S'}\right\rangle\left|\bar{0} + x_{S/S'}\right\rangle\big)
\end{aligned}
$$

$\square$

2 Toric Encoded Circuits

Definition 2.1 (Troic Encoded Circuit)**.** Let C be a quantum circuit. We say that C is *Toric encoded* if the following holds:

1. There is a Toric code $\mathcal{C}$ such that the qubits of $\mathcal{C}$ can be partitioned into registers $R_1, \ldots, R_k$, and each of the registers is encoded by $\mathcal{C}$.

2. The gates in C with even time coordinates are sweep decoding gates.

For a set of faults $\mathcal{E}$ the gates in $C[\mathcal{E}]$ can be decomposed to cycles, of the following form:

$$C = D_l P_l \Lambda_l, .., D_1 P_1 \Lambda_1$$

Definition 2.2 (Base graph G_o)**.** Let C be a quantum circuit, at depth d, with registers $R_1, \ldots, R_k$, all of which are encoded by Toric codes with the same parameters (i.e., dimension, length, etc.). Let's denote by $\mathcal{G}_1, .., \mathcal{G}_k$ the Toric complexes that induce the codes. We define the base graph of C, denoted by $G_o = (V_o, E_o)$ as follows:

1. The vertices V_o are tuples, where the first coordinates correspond to a vertex in the union over the vertices in $\mathcal{G}_1, \ldots, \mathcal{G}_k$, and the second coordinate is associated with a time. Namely:

$$V_o := \bigsqcup_i V(\mathcal{G}_i) \times [d]$$

2. The edges E_o are derived from the original edges in $\mathcal{G}_1, \ldots, \mathcal{G}_k$, with the addition that any two vertices in preceding times t and $t+1$ that support qubits (faces) such that C acts non-trivially on those qubits at time t are also connected by an edge. Namely:

$$\begin{aligned} E_o := \Big\{ \{(v,t),(u,t')\} :\ & \text{if} \\ & \text{either there exists } i \text{ such } \{v,u\} \in \mathcal{G}_i \text{ and } t' = t, t+1 \\ & \text{or there exists } u_i = (g_i, l_i) \in C \text{ such } v, u \in l_{i,2} \text{ and } t = l_{i,1}, t' = l_{i,1} + 1 \Big\} \end{aligned}$$

[COMMENT] Here we should add that a vertex v belongs to location l if there is a qubit f that v belongs to its associated face.

Definition 2.3 (Time Graph of $C(\mathcal{E})$)**.** For a quantum circuit C and set of faults $\mathcal{E}$, let us denote the base graph of $C[\mathcal{E}]$ by G_o. For each time t, denote the deviation induced by $\mathcal{E}$ at time t by $\mathcal{D}_t$, so $\partial \mathcal{D}_t$ is the syndrome that the qubits in the deviation admit. We define the history graph $\mathcal{H} = (V_\mathcal{H}, E_\mathcal{H})$ as follows:

1. The vertices $V_\mathcal{H}$ are taken to be the vertices of G_o, namely tuples $(v,t) \in V_o$, that satisfy that v belongs to a face associated with a check, which, at time t, belongs to the $\partial \mathcal{D}_t$. Namely:

$$\left\{ v_t : v_t = (v,t) \in V_o \text{ and there exists a check } f \in C \text{ such } v \in f \in \partial \mathcal{D}_t \right\}$$

2. The edges $E_\mathcal{H}$ is partitioned into two types $A_\mathcal{H}$ and $F_\mathcal{H}$, where $A_\mathcal{H}$ are the edges in E_o restricted to $V_\mathcal{H}$ and $F_\mathcal{H}$ are defined to be the . Namely:

$$\begin{aligned} A_\mathcal{H} &= \left\{ e = \{u,v\} \in E_o : u, v \in V_\mathcal{H} \text{ and } T(v) = T(u) \pm 1 \right\} \\ F_\mathcal{H} &= \left\{ e = \{u,v\} \in E_o :\ \text{exists } w \in V_\mathcal{H} \text{ such } \{v,w\}, \{u,w\} \in A_\mathcal{H} \right\} \end{aligned}$$

By definition of the construction, $G_\mathcal{H}$ is a time-graph.

Definition 2.4 (**Size**()-Function)**.** Let S be a subset of vertices, we define the *size* of S to be:

$$\mathbf{Size}(S) := \sum_{i}^{d+1} \max_{v \in S} L_i(v)$$

Claim 2.5. Let $\mathbf{S} = \{S_1, S_2, .., S_k\}$ where $S_i \subset \mathbb{R}^d$, introduce an edge between S_i and S_j iff $S_i \cap S_j \neq \emptyset$. Assume that $\mathbf{S}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^{k} S_i) \leq \sum_{i=1}^{k} \mathbf{Size}(S_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$\begin{aligned}
\mathbf{Size}(R_1 \cup R_2) &= \sum_{i}^{d+1} \max a_{i,1}, a_{i,2} = \sum_{i}^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\
&\leq \sum_{i}^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\
&= \mathbf{Size}(R_1) + \mathbf{Size}(R_2)
\end{aligned}$$

□

Definition 2.6 (Size Invariant Action)**.** [**COMMENT**] Here we classify both transversal gates, and rotation on the Toric. An action g is be said size invariant action if, for any assignment z, it holds that $\mathbf{Size}(gz) = \mathbf{Size}(z)$.

Claim 2.7. The sweep rule is Size ξ-shrinking.

Proof. □

[**COMMENT**] Eventually, we might not used the projected graph, and then we will split to history of the slices lemma for 3 parts. The first is when time $= 3t$, in that we apply an invariant action so.